CAGT

Recall AI
Use-Case Specification

Version 2.0

Revision History

Date	Version	Description	Author
11/007/26	0.1	Draft version	Nguyen Minh Phat
12/007/26	1.0	Review and update	Nguyen The Quan
26/07/2026	2.0	Revised the five existing use cases for precision and consistency with current requirements (text-first Interview with optional voice layer; per-concept prerequisite Traceback using a weighted mastery_score; Focus Session updates study statistics, not mastery_score; single defined behaviour for the Concept Graph empty-search case). Added three new use cases: Dashboard Overview, Study Material Management, and Session History.	Nguyen The Quan

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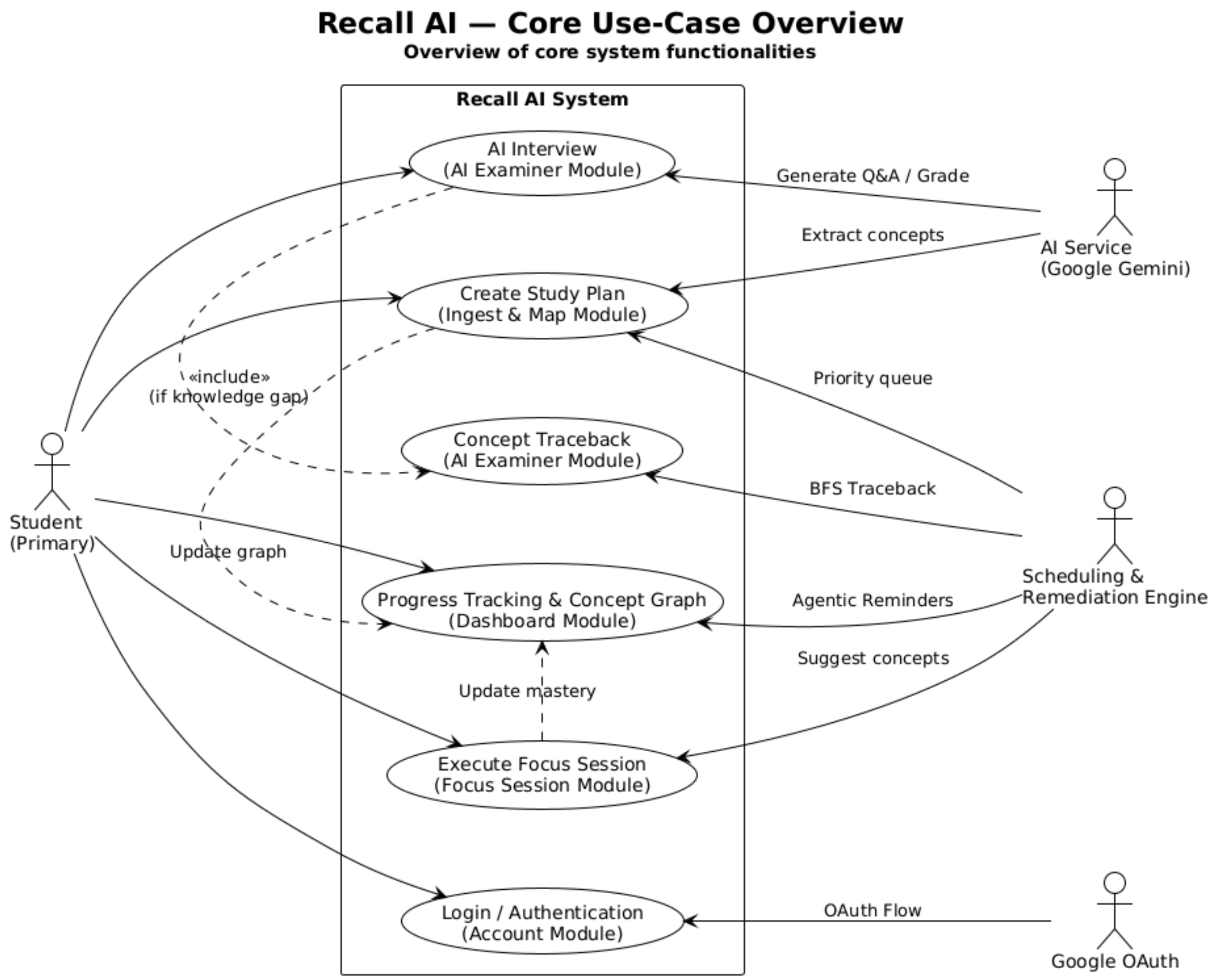
1. Use-case Model



2. Use-case Specifications

2.1 Use-case: Create Study Plan

Use case Name	Create Study Plan
Brief description	This use case allows a Student to create a new study plan by uploading study materials (text, images, PDF). The system will automatically call the AI Service to extract concepts and build a prerequisite relationship graph between them. After the user reviews and customizes the graph, the `Scheduling & Remediation Engine` will initialize an initial study schedule based on the concept graph and the set deadline
Actors	- **Student** - **AI Service** - **Scheduling & Remediation Engine (SRE)**

Basic Flow	1. The **Student** selects the “Create study plan” function on the system the system interface (Dashboard) 2. The **System** displays a form requesting study plan information, including: subject/plan name, deadline, and a document upload area. 3. The **Student** fill in the necessary information, uploads the study material (supporting Text, Image, PDF formats), and clicks the “Create” button. 4. The **System** receives the data and sends the document to the **AI Service** with a request to extract the list of concepts (with difficulty) and prerequisite relationships. 5. The **AI Service** analyzes the document and return the result to the system in a fixed-structure JSON format. 6. The **System** receives the JSON data and proceeds to validate the graph’s validity (Validate DAG - Directed Acyclic Graph) to ensure there are no cycles between concepts. 7. The **System** displays the newly extracted visual graph (concept map) on the screen and asks the **Student** to review it (This step includes the Use case *Interact with Plan Graph*. 8. The **Student** reviews, can add/remove node/edges if necessary. Every time a new edge is added, the **System** performs a real-time cycle check (DAG validation). If a cycle is detected, the edit is rejected with a warning. After all edits, the Student confirm the completion of the graph. 9. The **System** saves the plan information, concepts (`concepts`), and graph relationships (`concept_edges`) into the database. 10. The **System** calls the **Scheduling & Remediation Engine (SRE)** to generate an initial study schedule based on the finalized concept graph and the Student’s deadline. 11. The **System** displays a success message for plan creation and redicrects the Study to the Dashboard to view progress.
Alternative Flows	**Alternative flow 1: Invalid input material** 1. From #3 of the basic flow, if the uploaded document is not in a supported format, is blurry/unreadable (for images), or exceed ithe allowed file size, the system display an error message directly on the form and asks the Student to reselect the document. 2. The Student reselects the document. 3. Continue step #3 in the basic flow **Alternative flow 2: AI Service return invalid JSON format** 1. From #5 of the basic flow, if the AI Service returns free text or malformed JSON, the system automatically retries the request up to a maximum number of attempts. 2. If the maximum number of retries is reached and it still fails, the system reports an automatic extraction error and allows the Student to manually input concepts. 3. Continue step #8 in the basic flow (Sudent manually inputs concepts)

	Alternative flow 3: Generated graph constraints cycles 1. From #6 of the basic flow, if the system detects that the extracted graph contains cycles (is not a DAG), it automatically removes the edge causing the cycle and log a warning. 2. Continue step #7 in the basic flow for the Student to manually check. **Alternative flow 4: AI Service encounters an error or is overloaded** 1. From #5 of the basic flow, if the AI Service does not respond (timeout) or the system runs out of API quota, the system displays the error: “AI analysis service is interrupted, please try again later” 2. The use case terminates **Alternatice flow 5: Student cancels the plan creation** 1. At any point before step #8 of the basic flow, if the Student clicks “Cancel” or leaves the page, the system deletes the temporary unsaved data. 2. The use case terminates.
Pre-conditions	- The Student has successfully logged into the Recall AI system. - The AI Service is in a normal operating state, and the system has sufficient API quota/rate limit
Post-conditions	- A new Study Plan record is successfully saved in the database. - The concepts and corresponding prerequisite relationships of the plan are fully stored as a Directed Acyclic Graph (DAG) - An initial study schedule for the plan has created and is ready for study sessions (Focus Session/Interview).

2.2 Use-case: Execute Focus Session

Use case Name	Execute Focus Session
Brief desciption	This use case describes the process of a Student executing a time-managed study session (e.g., Pomodoro). During the session, the Student can view the original document (PDF) side-by-side and take quick notes related to the concepts being studied. Upon completion, the system saves the session record (time studied) and interacts with the Scheduling & Remediation Engine (SRE) to update the progress (mastery score), preparing for the subsequent knowledge verification step.
Actor	- Student - Scheduling & Remediation Engine (SRE)
Basic Flow	**1. Student** requests to start a new study session from the application interface (Dashboard or subject details screen). **2. System** displays the setup screen, asking the Student to select a concept

	to review and configure the study time (default to Pomodoro). **3.** **Student** selects the desired concept and confirms to start. **4.** **System** displays the main session interface, including the countdown timer, side-by-side original PDF document view, and the note-taking tool **5.** **Student** clicks the "Start" button. **6.** **System** starts the countdown timer. **7.** **Student** read the document, studies, and enters quick notes linked to the concept. **8.** **System** automatically auto-saves the Student's notes. 9. When the timer finishes counting down (or Student manually click to end), the **Student** selects "End session". **10.** **System** saves the entire session record (total time, concepts, notes) and send the information to the SRE. **11.** **SRE (Actor)** receives the information and updates the concept's study statistics (total study time, session count). It does NOT modify the mastery_score, which is set only by the AI Examiner (Execute Interview Session). **12.** **System** displays the session completion screen with a summary of the results, providing subsequent options (e.g., take a break or move to `AI Examiner`).
Alternative Flows	**Alternative flow 1: View and select Concept Recommendations (Extension)** 1. From step #2 of the basic flow, if the Student selects the "View Recommendations" feature. 2. System sends a request to the SRE. SRE calculates and provides a list of priority concepts (based on weak mastery or upcoming deadlines). System displays the list of recommendations to the Student and Student selects a concept from the recommended list. 3. Continue step #3. **Alternative flow 2: Enable Strict Mode (Extension)** 1. From step #4 or #6 of the basic flow, if the Student selects to enable "Strict Mode". 2. System activates the mechanism to block/restrict distracting websites or tabs. 3. Continue step #4 (or #6) **Alternative flow 3: Pause and Resume** 1. From step #6 of the basic flow, if the Student clicks the "Pause" button on the timer. 2. System pauses the countdown timer and saves the current state. The Student clicks "Resume". System resumes the countdown timer. 3. Continue step #6

	Alternative flow 4: Cancel the session midway 1. From step #5 to step #8 of the basic flow, if the Student selects "Cancle session" 2. System displays a dialog asking to confirm the cancellation. The Student confirms the cancellation. System closes the session, does not record the study time (or saves it "Cancelled" status), and redirects the user to previous interface. 3. The use case terminates
Pre-conditions	- The Student has successfully logged into the Recall AI system. - The Student has created a Study Plan, and the system already has a list of concepts along with their corresponding prerequisite graph.
Post-conditions	- The study duration and session status are successfully saved to the system history. - The student's quick notes are saved and correctly linked to the selected concepts. - The SRE record the session completion data to update mastery parameters (study statistics)

2.3 Use-case: Execute Interview Session

Use case Name	Execute Interview Session (Multi-turn Q&A)
Brief desciption	This use case allows a Student to take a knowledge test in the form of a multi-turn Q&A interview. The system use AI to act as an examiner, generating questions and grading answers directly based on the original materials uploaded by the Student. The workflow is controlled by a state machine that limits the number of Q&A turns, helping to classify the actual level of comprehension and automatically triggering a prerequisite traceback if it detects the Student has gaps in foundational knowledge.
Actor	- Student - AI Service
Basic Flow	**1. Student** starts the interview session from the queue session of concepts to review. **2. System** loads the highest priority concept, sets the limit for Q&A turns (e.g., 3 turns), and sends a request (`generate_question`) to the **AI Service** **3. AI Service** generates an interview question (generate_question) based on the rubric and the original uploaded material.

	4. System displays the generated question to the Student as text. An optional voice layer (text-to-speech playback of the question, with the text hidden until the Student taps "Show question") is available as an extension — see Alternative flow 5. **5. Student** types the answer to the question. Optionally, the Student may answer by voice through the device's microphone (see Alternative flow 5). **6. System** receives the Student's typed answer. When voice input is used, the spoken answer is transcribed to text (Speech-to-Text) so the Student can review and edit it before submitting. **7. System** sends the sends the text answer to the **AI Service** for comparison and grading (`grade_answer`). **8. AI Service** analyzes and returns the score, feedback, and the level of understanding classification **9. System** processes the AI's feedback: - If the answer shows **deep understanding**, the system asks the AI to create more profound application question (loops back to step 3) - If the answer is **superficial**, the system asks the AI to pose a follow-up question forcing the Student to explain more clearly (loops back to step 3) - If the answer is **wrong/missing knowledge**, the grading feedback (from grade_answer) explains the mistake and the current concept ends early. Whether a prerequisite Traceback is triggered is decided after the concept's final mastery_score is computed (see 2.5 Concept Traceback) **10. System** saves the evaluation result of the current concept after reaching the maximum number of turns or finishing the processing loop in step 9. **11. System** repeats the process from Step 2 for the next concept in the queue until the list is completed. **12. System** announces the end of the interview session via voice and automatically redirects the Student to the End-of-Session Summary screen.
Alternative Flows	**Alternative flow 1: AI Service Timeout or Error (AI Fail/Timeout)** 1. From #3 or #7 of the basic flow, if the AI Service responds to slowly, runs out of quota, or reports an error. 2. The system log the error and automatically Fallback. The system stop calling the AI, switches to a static flashcard interface (using pre-generated questions). The Student self-evaluates as correct/incorrect. 3. Continue step #11 **Alternative flow 2: Student Pause the interview session** 1. From #5 of the basic flow, if the Student selects the "Pause" option. 2. The system saves the current state (including conversation history, remaining turns) and exits the interface. (When the Student returns, the system will replay the question from Step #4) 3. The use case terminates

	Alternative flow 3: Student Skips a concept 1. From #4 or #5 of the basic flow, if the Student selects the “Skip” option. 2. The system immediately cancels the remaining returns and skips grading. 3. Continue step #10 **Alternative flow 4: Student Appeals the grading result** 1. From #9 of the basic flow, if after hearing the AI’s feedback, the Student disagrees and selects “Appeals” 2. The system pauses the voice conversation flow, displays a form to enter the reason for the appeal. The concept’s result is temporarily saved with and “Appealing” flag 3. Continue step #10
Pre-conditions	- The Student has logged in and completed the interview session configuration step - The list of concepts to review has been loaded into the queue by the system. - The system has a stable network connection to the AI service. - The Student has granted microphone access to the browser or device (only required when the optional voice input mode is used).
Post-conditions	- The comprehension score (`mastery_score`) of each concept tested in the session is fully updated in the database (`concepts` table). - If knowledge gaps are found, the foundational (prerequisite) concepts have been automatically inserted into the queue for the next study session via the Concept Graph Engine’s algorithm. - The entire conversation history (text transcribed from speed and/or audio data) is successfully stored to serve as display data for the summary screen and future learning processes.

2.4 Use-case: Concept Graph Interaction

Use case Name	Concept Graph Interaction
Brief desciption	This use case allows Students to visually interact with the Concept Graph of their courses or study plans. By interacting through basic actions such as zoom, pan, hover tooltip, as well as searching and filtering concepts by their mastery score, Students can quickly grasp the overall picture of their knowledge gaps and the prerequisite relationships between concepts.
Actor	- **Student**

Basic Flow	1. **Student** selects the Concept Graph feature from the main Dashboard or through the main menu. 2. **System** loads and displays the network diagram of concepts. The nodes (concepts) are colored based on the student's level of mastery (according to mastery_score). 3. **Student** per forms actions on the graph: - **Zoom**: Zoom in or out to focus on a specific cluster of concepts or to view the entire graph. - **Pan**: Drag and move the viewport around the graph space. - **Hover tooltip**: Hover the mouse over a concept node. The system displays a tooltip with a brief summary, including the concept's name and its current mastery_score. 4. **Student** uses the toolbar to search and filter: - Enter a keyword in the search box to filter concepts by name. - Select a mastery level filter (e.g., weak, strong - corresponding to specific mastery_score ranges). 5. **System** automatically highlights the concepts that meet the search/filter criteria and dims the remaining nodes. 6. **Student** clicks on a concept node on the graph. 7. **System** displays a detail panel (sidebar/modal) for the selected concept, which includes: - Study history (History) of that concept across previous learning sessions. - A list of prerequisite concepts (Prerequisites) that must be mastered beforehand.
Alternative Flows	**Alternative flow 1: No concepts found when filtering/searching** 1. From step #5 of the basic flow, if the keyword or filter conditions do not match any concept in the graph. 2. The system displays the message "No matching concepts found" and dims all nodes in the graph. 3. Continue step #4. **Alternative flow 2: Error loading graph data** 1. From step #2 of the basic flow, if an error occurs while retrieving graph data from the database (network error or server error). 2. The system displays the error message "Unable to load the concept graph. Please try again later" and provides a "Try again" button so the student can retry the connection. 3. The use case terminates.
Pre-conditions	- The Student has successfully logged into the Recall AI system. - The Student has at least one study plan successfully extracted that contains a concept graph.

Post-conditions	- The system undergoes no data changes (this is a read-only task). The Student gains an overview of their knowledge mastery and the location of their weaknesses within the knowledge chain based on the visual graph.

2.5 Use-case: Concept Traceback

Use case Name	Concept Traceback
Brief desciption	This use case describes the process where the system automatically traces the foundational knowledge gaps of a student when they answer incorrectly or receive an evaluation score below the threshold for a specific concept. The system utilizes a reverse Breadth-First Search (BFS) algorithm on the Concept Graph to identify weak prerequisites. This process runs entirely on software logic (purely Agentic, no AI API calls) to produce a list of concepts to review, allowing the student to verify and insert them into their upcoming study schedule.
Actor	- **Scheduling & Remediation Engine (SRE)** - **Student**
Basic Flow	**1.** After a concept has been fully assessed in the Interview Session (all of its Q&A turns are finished), its mastery_score is computed as a weighted average of the turn scores (later turns weigh more). If concept \`C\` has a mastery_score < threshold, the **SRE** automatically triggers the Concept Traceback feature. **2.** The **SRE** initializes an empty queue Q. **3.** The **SRE** queries the database (concept_edges) to retrieve all direct prerequisite concepts of C and enqueues them into Q. **4.** The **SRE** iterates through the elements in queue Q. The traversal is constrained to a maximum depth of 2 levels (max_depth = 2) from C to prevent the subsequent study session from becoming excessively long. **5.** For each prerequisite concept P dequeued from Q, the **SRE** checks the score history of P. The system determines if P has never been tested or if its mastery_score(P) < threshold. **6.** If the condition in step #5 is met, the **SRE** inserts concept P at the front of the queue for the next study/review session, prioritizing learning P before returning to review C. Simultaneously, the prerequisite concepts of P (if the 2-level depth limit has not been exceeded) are enqueued to the end of Q. **7.** After the traversal loop is complete, the system displays the newly found list of weak prerequisite concepts (prereqs) on the UI. **8.** The student reviews the list of lacking foundational concepts and can then

	choose to confirm and add them to their study schedule, or skip one or more concepts. **9.** The SRE saves the new schedule configuration based on the student's feedback and terminates the flow.
Alternative Flows	**Alternative flow 1: Prerequisite Concept is Mastered** 1. From step #5 of the basic flow, if the SRE detects that the prerequisite concept P has been previously tested and has a mastery_score(P) >= threshold. 2. The SRE skips P (since this knowledge is already mastered and requires no review) and dequeues the next element in Q. The system will not traverse deeper into the prerequisites of this P. 3. Continue step #4. **Alternative flow 2: No Prerequisites Found to Review** 1. From step #4 of the basic flow, after completing the traversal of queue Q (queue is empty or max_depth reached), if the SRE finds no prerequisite concepts that are weak or untested. 2. The SRE treats concept C via the normal spaced repetition mechanism: it schedules a review for C after X days instead of an immediate review. 3. Continue step #9. **Alternative flow 3: Student Skips Foundational Concepts** 1. From step #8 of the basic flow, if the student decides to skip all or some of the foundational concepts proposed by the system. 2. The SRE removes the skipped concepts from the priority queue of the next session. The remaining concepts (if any) are still prioritized; for concept C, the system will schedule it for review as normal if all prerequisites are skipped. 3. Continue step #9.
Pre-conditions	- The concept graph for the study material/subject has been successfully created, including the concepts and the prerequisite relationships between them (concept_edges). - The student is in the middle of or has just completed an Interview Session. - The system detects a concept (referred to as concept C) that was answered incorrectly or has a mastery_score lower than the specified threshold (e.g., < 0.6).
Post-conditions	- The weak prerequisite concepts are successfully identified by the system. - The student's study schedule is updated: Foundational concepts needing reinforcement are inserted at the front of the priority queue for the next study session (to be studied before concept C).

	- Information about the weak prerequisites is stored so it can be displayed in detail in the End-of-Session Summary.

2.6 Use-case: View Dashboard Overview

Use case Name	View Dashboard Overview
Brief description	This use case describes how a Student obtains a quick, consolidated overview of their overall study progress on the Dashboard, which is the default screen shown after login. The Dashboard aggregates the Student's active study plans and their progress, upcoming deadlines, the concepts recommended for review today (computed by the Scheduling & Remediation Engine), a colour-coded mini concept graph, and quick study statistics.
Actors	- Student - Scheduling & Remediation Engine (SRE)
Basic Flow	1. The Student logs in; the System opens the Dashboard as the default screen. 2. The System loads and displays: (a) active study plans with progress %, deadline and days remaining; (b) an upcoming-deadline timeline; (c) the "Concepts to review today" list produced by the SRE, each with a short reason; (d) a mini concept graph whose nodes are coloured by mastery_score; (e) quick statistics such as total study time this week and number of Interview sessions completed. 3. The Student clicks a recommended concept to start a Focus Session or an Interview for it. 4. The Student may open the full Concept Graph (see 2.4) or a plan's detail from the Dashboard.
Alternative Flows	**Alternative flow 1: No study plan yet** 1. From step #2, if the Student has no study plan, the System shows an onboarding state with a "Create your first plan" call-to-action. 2. The use case continues when a plan is created. **Alternative flow 2: All plans past their deadline** 1. From step #2, if every plan is past its deadline, the System highlights this and suggests creating a new plan. **Alternative flow 3: Error loading dashboard data** 1. From step #2, if the data cannot be loaded (network or server error), the System shows an error message and a "Retry" button. The use case terminates.
Pre-conditions	- The Student has successfully logged into the Recall AI system.
Post-conditions	- No data is changed (this is a read-only task). - The Student has an up-to-date overview of their progress and of the concepts recommended for today.

2.7 Use-case: Manage Study Materials and Plans

Use case Name	Manage Study Materials and Plans
Brief description	This use case allows a Student to manage their existing study plans and the materials behind them: view the list of plans, open a plan to inspect its concepts and graph, rename / archive / delete a plan, and re-analyse a plan when its source material changes. Re-analysis re-runs concept extraction and updates the

	concept graph while preserving existing mastery history for concepts that still match.
Actors	- Student - AI Service (Google Gemini) — for re-analysis - Scheduling & Remediation Engine (SRE) — reschedules after a graph change
Basic Flow	1. The Student opens “My Plans” from the Dashboard or the main menu. 2. The System displays the list of the Student’s study plans with status (active / archived), progress and deadline. 3. The Student selects a plan to view its details (concepts, concept graph, schedule). 4. The Student performs a management action: rename the plan, archive it, delete it, or re-analyse it after replacing or adding material. 5. For rename / archive / delete, the System updates the plan record and confirms the change. 6. For re-analyse, the System sends the updated material to the AI Service to re-extract concepts and prerequisite relationships, validates the resulting graph (DAG), and asks the Student to review and confirm the updated graph (see 2.4). Concepts that still match keep their mastery history; concepts that disappear are deprecated. 7. After a graph change, the System calls the SRE to update the study schedule. 8. The System confirms the change and returns the Student to the plan detail.
Alternative Flows	**Alternative flow 1: Delete a plan** 1. From step #4, deleting a plan requires an explicit confirmation. On confirm, the plan and all of its related data (concepts, sessions, review-queue items) are permanently removed. **Alternative flow 2: Archive instead of delete** 1. From step #4, the Student archives a plan. It is hidden from active views and excluded from the “today” recommendations, but all of its data is retained and can be restored. **Alternative flow 3: Re-analysis fails** 1. From step #6, if the AI Service errors or returns invalid JSON after the maximum number of retries, the System keeps the previous graph unchanged, shows an error, and lets the Student retry later. Continue step #3.
Pre-conditions	- The Student is logged in and has at least one study plan.
Post-conditions	- The selected management action is persisted in the database. - For re-analysis, the plan’s concepts, graph and schedule are updated while valid mastery history is preserved. This use case never resets mastery_score.

2.8 Use-case: View Session History

Use case Name	View Session History
Brief description	This use case lets a Student review their past learning sessions over time — both Focus Sessions and Interview Sessions — in order to track study habits and progress. The Student can browse a chronological list, open a session to see its details, and view simple aggregate statistics.

Actors	- Student
Basic Flow	1. The Student opens “History & Progress” from the main menu. 2. The System displays two tabs — Interview Sessions and Focus Sessions — each listed most-recent-first with date, concepts, and a short summary (Interview: average score; Focus: total study time). 3. The Student selects a session to view its details: • For an Interview Session: each Q&A turn (question, the Student’s answer, the AI feedback, the score and the verdict) and the mastery_score of each concept before and after the session. • For a Focus Session: the total study time and the concepts studied. 4. The System shows aggregate statistics, for example total study time this week, number of sessions, and the mastery trend per concept over time.
Alternative Flows	**Alternative flow 1: No sessions yet** 1. From step #2, if the Student has no completed session, the System shows an empty state with a call-to-action to start the first session. **Alternative flow 2: Interrupted or abandoned session** 1. From step #3, an interrupted session is still listed but clearly marked as incomplete, showing only the data captured before it stopped.
Pre-conditions	- The Student has successfully logged into the Recall AI system. - To see content, the Student has at least one completed Focus or Interview session.
Post-conditions	- No data is changed (this is a read-only task). - The Student has reviewed their learning history and progress trends.